

Nafees S

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Education

B.M.S. College of Engineering — *B.E. in Computer Science & Engineering* CGPA: 8.21 *Expected June 2027*

Narayana PU College — *12th, Karnataka State Board* **94.33%** *May 2023*

Technical Skills

Languages: Python, Java, C, JavaScript, TypeScript, Swift, SQL

Frameworks & Databases: FastAPI, React, Tailwind CSS, SwiftUI, PostgreSQL, SQLAlchemy, Alembic, Redis, Celery, Vite

AI / ML: LLMs, Prompt Engineering, Claude & Gemini APIs, RAG, PPO Reinforcement Learning, Deep Learning, NLP, Multimodal Fusion (CLIP), LangChain, ChromaDB

Tools: Git/GitHub, Docker, AWS EC2, GitHub OAuth/Webhooks, Xcode, Figma, Jupyter Notebook, VS Code

Coursework: Machine Learning, Linear Algebra, Probability & Statistics, Data Structures & Algorithms, Advanced Data Structures

Projects & Research

Liffy — Open-Source AI Peer Code Review Tool (Project Lead) — FastAPI, [Source Code](#) 
Celery, PostgreSQL, Redis, React, TypeScript, ChromaDB, LangChain

- Led the project as primary owner (top committer by a wide margin), defining a layer-ownership model with a collaborator across backend, frontend, and LLM-pipeline work
- Directed a self-hosted platform ingesting GitHub PR webhooks and generating line-anchored feedback via a RAG pipeline (ChromaDB + LangChain) grounded in the repo's code, with an async Celery + Redis job layer, a provider-agnostic LLM interface (Ollama, Gemini, Anthropic, Claude Code/Codex), and one-command bootstrap scripts for macOS/Windows

Askcal — AI-Powered Daily Scheduler — FastAPI, Claude, PostgreSQL, SwiftUI, [Source Code](#) 
Google OAuth

- Built a regret-ranked inbox classifier that never asks the model for a score: it extracts stakes, sender, deadline and task signals, and deterministic arithmetic ranks each mail 0–100 behind confidence floors
- Shipped the SwiftUI iOS/iPadOS app over FastAPI/Postgres, scheduling tasks around real Google Calendar busy blocks; 282 backend tests in CI, refresh tokens encrypted at rest under rotatable keys

Interpretable PPO-Based Adaptive Traffic Signal Control — Python, SUMO, [Source Code](#) 
TraCI, RL

- Implemented a PPO agent with curriculum learning and attention-based feature attribution to address the black-box challenge in AI-driven infrastructure, using real-time bidirectional SUMO–TraCI communication with constrained emergency preemption and priority queuing

Tiket — Full-Stack Ticket Booking System — SwiftUI, FastAPI, PostgreSQL, [Source Code](#) 
AWS EC2, JWT

- Designed and deployed a full-stack ticket booking system with concurrency-safe seat allocation (PostgreSQL `SELECT FOR UPDATE`), JWT auth, and AWS EC2 deployment — load tested at 12,441 concurrent requests with zero double bookings; built the SwiftUI frontend with real-time seat selection and dynamic barcode generation

Leadership

Senior Coordinator, Phase Shift 2025 — *BMSCE* *2024 – 2025*

- Co-led logistics for BMSCE's annual technical fest; previously Junior Coordinator for Utsav 2025, managing design team and on-ground event operations